SQL AGGREGATE FUNCTIONS & OPERATORS

EXERCISE 2

Q1) SELECT DISTINCT department
FROM Students;

▲ TABLE Students

department
IT
HR
FINANCE

Q2) SELECT department,
AVG (age) As average_age_student
FROM Students
Group By department;

▲ TABLE Student

department	Average_age
IT	20
HR	22
Finance	23

Q3) SELECT department,
COUNT (*) As Student_count
FROM Students
Group By department
HAVING COUNT(*) > 1;

▲ TABLE Students

department	Student_Count
IT	2
HR	2

Powered by CamScanner

Q4) SELECT student_id,
name,
age
FROM students
WHERE age BETWEEN 21 AND 23;

TABLE Students

Student_id	name	age
2	Bob	22
3	Charlie	21
4	Diana	23
5	Eve	22

Q5) SELECT *
FROM students
WHERE students IN ('IT', 'HR')
AND ages > 21;

TABLE Students

Student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Q6) SELECT department,
SUM (credits) AS total_credits_per_department
FROM courses
Group By department
HAVING SUM (credits) > 5;

department	total_credits
IT	11

4 + 4 + 3 = 11

Powered by CamScanner

Q7) SELECT *
FROM courses
WHERE credits != 4;

TABLE students

course-id	course name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	statistics	HR	3

Q8) SELECT course_id,
course_name,
credits
FROM courses
ORDER BY 'credits' DESC,
LIMIT 3;

TABLE students

course-id	course-name	credits
103	Data Science	4
102	Python	4
101	SQL Basics	3

Q9) SELECT enrollment_id,
MAX (grade) As max_grade,
MIN (grade) As min_grade,
AVG (grade) As total_average_grade
FROM enrollments;

max_grade	min_grade	avg_grade
90	78	423

Avarage
85+78+90+88+82
= 423

Powered by CamScanner

Q10) SELECT Course_id,
COUNT(*) As total_enrollment_Count
FROM enrollments
Group BY Course_id;

TABLE enrollments

Course_Id	enrollment_Count
101	1
102	1
103	1
104	1
105	1

Q11) SELECT department,
SUM (salary) As total_salary
SUM (bonus) As total_bonus_per department
FROM Salaries
Group BY departments;

TABLE Salaries

department	total_Salary	total_bonus
IT	122 000	105 00
HR	109 000	7500
Finance	70000	6000

Q12) SELECT department,
AVG (salary) As total_AVG_Salary
FROM Salaries
GROUP BY department
HAVING AVG(Salary) > 55 000;

Salaries

department	AVG_Salary
IT	61 000
finance	70 000

Powered by CamScanner

Q13) SELECT employee_id,
name,
Salary bonus,
SUM (Salary + bonus) As total_Salary_Plus_bonus
FROM Salaries
WHERE ('salary + bonus') > 60 000;

TABLE Salaries

employee_id	name	Salary	bonus	total_Salary+bonus
1	Tom	60 000	5 000	65 000
3	Spike	70 000	6000	76 000
4	Tyke	62 000	5500	67 500

Q14) SELECT departments,
SUM (budget) As total_budget_per department
AVG (budget) As total_average_per_department
FROM Projects
Group BY departments ~~> 70000~~
HAVING AVG (budget) > 70 000;

TABLE Projects

department	total_budget	total_AVG_budget
IT	27000	135000
Finance	80 000	80 000

Q15) SELECT *
FROM projects
WHERE budgets BETWEEN 50 000 AND 120 000,
AND department NOT 'marketing';

TABLE Projects

Project_Id	Project_name	department	budget
1	AI App	IT	12 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000

Powered by CamScanner